

# A Generative AI-Enhanced Hybrid Framework for Reliable Long-Term Photovoltaic Forecasting

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**Institution:** University of Glasgow **Industry sponsor:** Scottish Power Energy Networks

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## Background & Industry Challenge

The global transition towards **net-zero energy systems** requires accurate, long-term forecasting of renewable generation to support investment, capacity planning, and operational optimisation. However, **current forecasting practices remain focused on ultra-short-term and short-term horizons** (minutes to days), relying heavily on recent weather records and accurate **Year-Ahead Meteorology (YAM)** data that are **not available for year-ahead planning**.

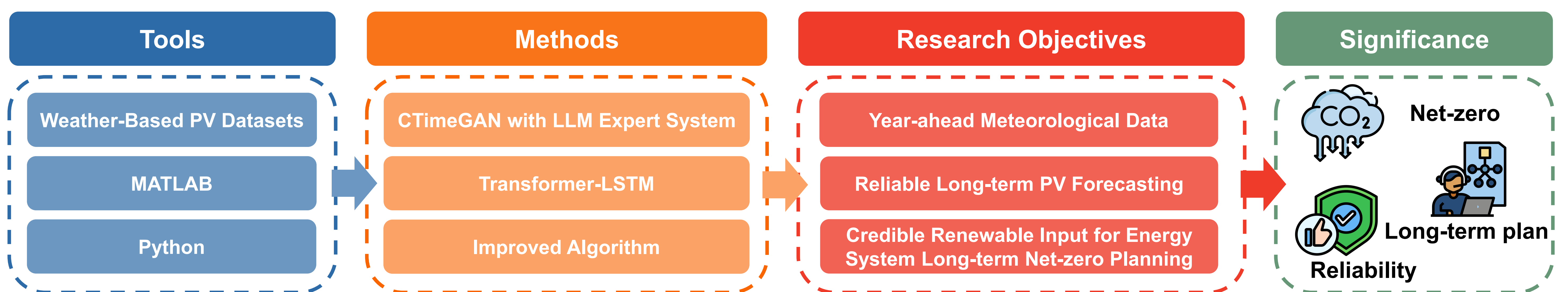
## Main Objectives

This research aims to overcome these challenges by developing an **intelligent, data-driven forecasting framework** that:

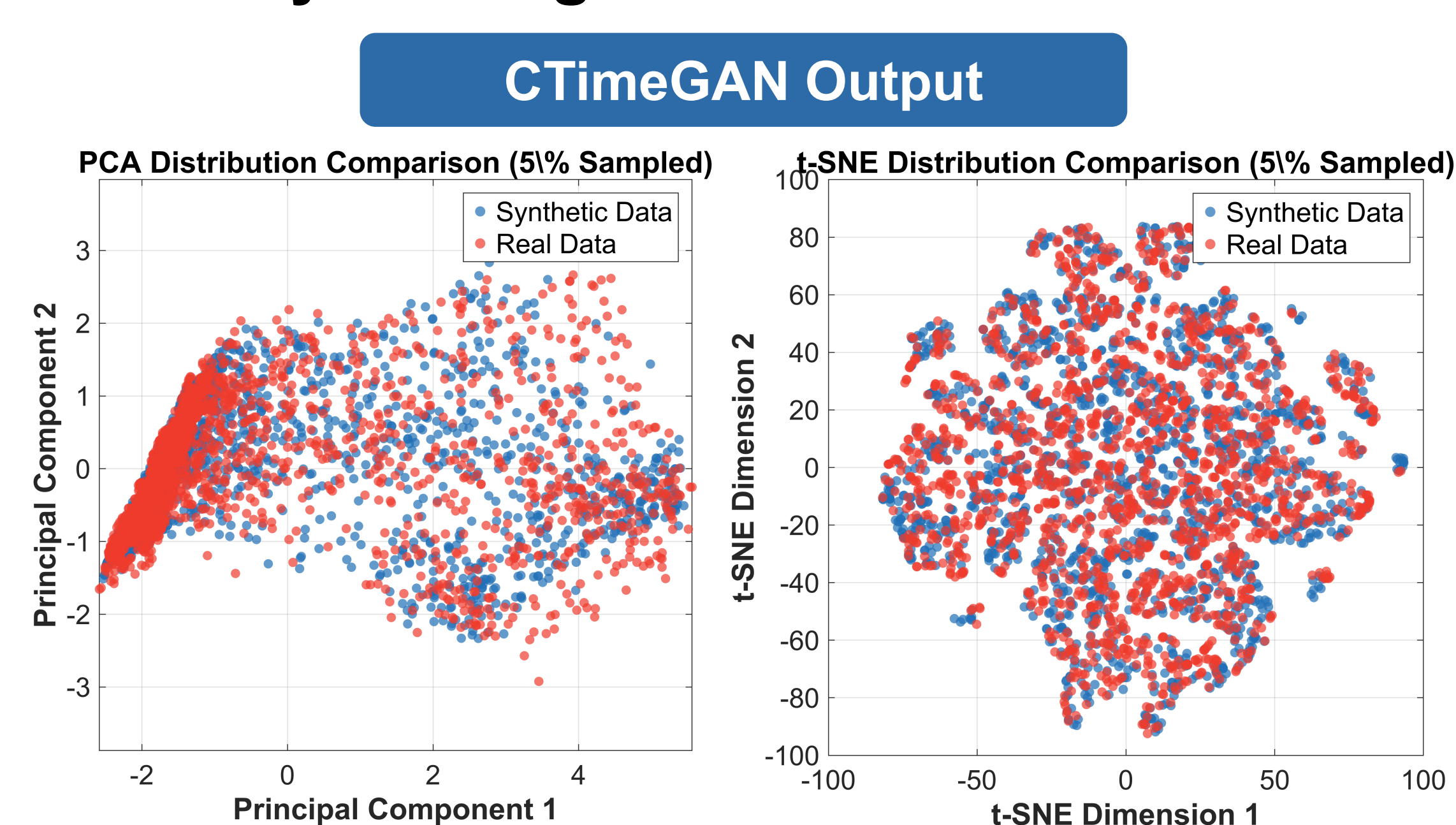
- Bridges the gap between short-term prediction and long-term planning, **enabling renewable forecasting on an annual scale**.
- Provides **high-quality renewable input scenarios** for multi-energy system optimisation, improving reliability, flexibility, and economic performance under future energy transitions.

This project has the potential to contribute to the **Scottish Government's Net Zero 2045 Target** and the **UK Energy Security & Net Zero Strategy** by enabling **long-term, uncertainty-aware energy planning** across diverse renewable and multi-energy systems, including electricity, heat, hydrogen, and transport, while promoting **AI-enhanced decision-making** for renewable integration and decarbonisation. Additionally, it promotes data innovation and digital collaboration with Scotland's clean energy ecosystem through collaboration between the ETP, academia, and industry partners.

## Research Framework

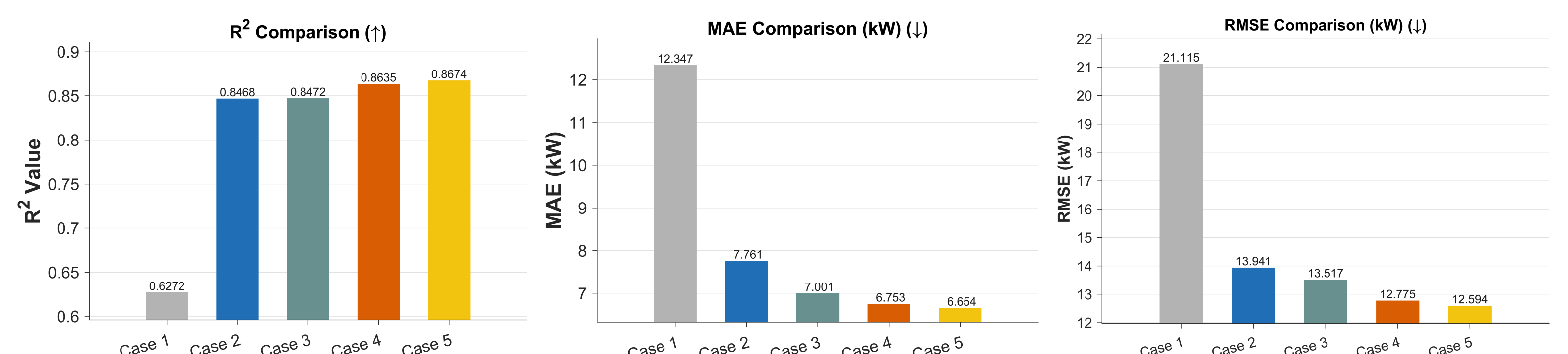


## Preliminary Findings



## Model Performance

| Case | Model                                         | R <sup>2</sup> | MAE (kW) | RMSE (kW) |
|------|-----------------------------------------------|----------------|----------|-----------|
| 1    | Regressive LSTM                               | 0.6272         | 12.3467  | 21.1147   |
| 2    | LSTM <b>CTimeGAN</b> <b>YAM</b>               | 0.8375         | 10.0288  | 16.8467   |
| 3    | Transformer-LSTM <b>CTimeGAN</b> <b>YAM</b>   | 0.8472         | 7.0009   | 13.5171   |
| 4    | Transformer-LSTM <b>Generative</b> <b>YAM</b> | 0.8635         | 6.7526   | 12.7755   |
| 5    | Proposed Model <b>Generative</b> <b>YAM</b>   | 0.8674         | 6.6545   | 12.5943   |



The proposed **Generative AI-Enhanced Hybrid Framework** effectively bridges the long-term PV forecasting gap by generating and refining future meteorological data. The **CTimeGAN + LLM Expert System** produces physically consistent, trend-aware Year-Ahead Meteorology (YAM), improving R<sup>2</sup> by 0.21 and reducing MAE and RMSE by over 35 %. After expert-guided correction and adaptive optimisation, the final **Generative** **YAM** **E-AEO-Transformer-LSTM** model achieved R<sup>2</sup> = 0.8674, MAE = 6.65 kW, and RMSE = 12.59 kW, **demonstrating superior accuracy and stability**.

Overall, the framework provides a robust, privacy-preserving, and scalable solution for long-horizon PV forecasting and future net-zero energy planning.

## Desired Outcomes & Impacts

This research aims to make a constructive contribution to **improving the reliability and practicality of long-term renewable forecasting**. The proposed generative AI-enhanced framework is designed to integrate physical consistency, statistical realism, and predictive adaptability, offering a flexible foundation for extending similar methods to other renewable sources and multi-energy systems. By combining generative modelling, intelligent optimisation, and deep learning, the study seeks to advance uncertainty-aware and data-efficient approaches within the wider energy research community.

Beyond academic significance, the work is expected to provide useful insights for energy planners and network operators by enabling **more informed long-term decision-making under net-zero transition scenarios**. Through collaboration within the ETP programme and with the support of Scottish Power Energy Networks, the project encourages responsible data sharing and digital innovation in Scotland's clean energy ecosystem, contributing constructively to the development of intelligent, resilient, and sustainable energy networks.